

Q2. Write the difference between Hub, Switch & Router in Tabular Format. Explain why router is the most intelligent & Hub is least intelligent device. (3 M)

Feature	Hub	Switch	Router
1. OSI Layer	Layer 1 – Physical	Layer 2 – Data Link	Layer 3 – Network
2. Address Used	No address	MAC Address	IP Address
3. Memory	No memory	Stores MAC Address Table	Stores Routing Table
4. Data Forwarding	Sends data to all ports	Sends data to the specific port	Sends data to the appropriate network
5. Main Function	Connects devices in a LAN	Connects devices and establishes LAN	Connects two or more different networks

Hub – Least Intelligent

- Hub works at Layer 1 (Physical Layer) and does not understand MAC or IP addresses.
- It broadcasts data to all ports, causing unnecessary traffic and collisions.

Router – Most Intelligent

- Router works at Layer 3 (Network Layer) and uses IP addresses to forward packets.
- It maintains a Routing Table and selects the best path between different networks, while also separating broadcast domains.

Q3. What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also Explain difference between CSMA/CD & CSMA/CA. (3 M)

Random Access Protocols:

1. Random Access Protocols are **MAC sub-layer protocols** used in shared communication mediums.
2. Every station has an **equal opportunity** to access the channel.
3. No station has a **permanently reserved channel**.
4. Whenever a station has data, it **competes for access** to the medium.
5. If more than one station transmits at the same time, a **collision** can occur.
6. Collision can **corrupt or destroy frames**.

Need of Random Access Protocols:

Random Access Protocols are needed to manage a **shared communication medium**.

They provide procedures for:

1. **When** a station can access the medium.
 2. What to do when the **medium is busy**.
 3. How to determine **transmission success or failure**.
 4. How to **handle collisions**.
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Categories of Random Access Protocols

Category A: ALOHA

ALOHA allows multiple stations to transmit data over a **shared channel**.

Types:

1. **Pure ALOHA**
2. **Slotted ALOHA**

1. Pure ALOHA

1. Station transmits whenever it has data; it **does not check whether the medium is free**.
2. After transmission, it waits for an **ACK**.
3. If ACK is not received within the time limit, it assumes a **collision occurred**.
4. It waits for a random **Back-off Time (Tb)**.
5. Then it retransmits the frame.

Vulnerable Time

$$VT = 2 \times T_{fr}$$

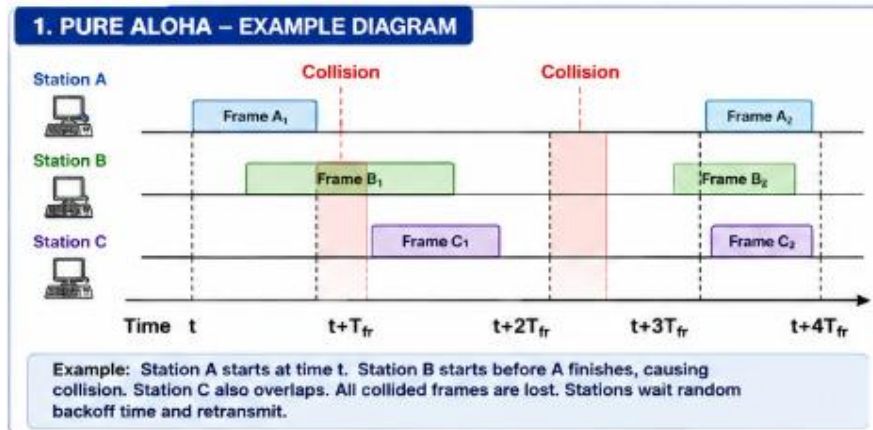
Where T_{fr} = Frame Transmission Time.

Throughput

$$S = G \times e^{-2G}$$

Maximum efficiency $\approx 18\%$

Example



2. Slotted ALOHA

1. The shared channel is divided into **discrete time slots**.
2. Stations can transmit only at the **beginning of a time slot**.
3. If a station misses the beginning of a slot, it waits for the **next slot**.
4. This reduces the probability of collisions compared to Pure ALOHA.

Vulnerable Time

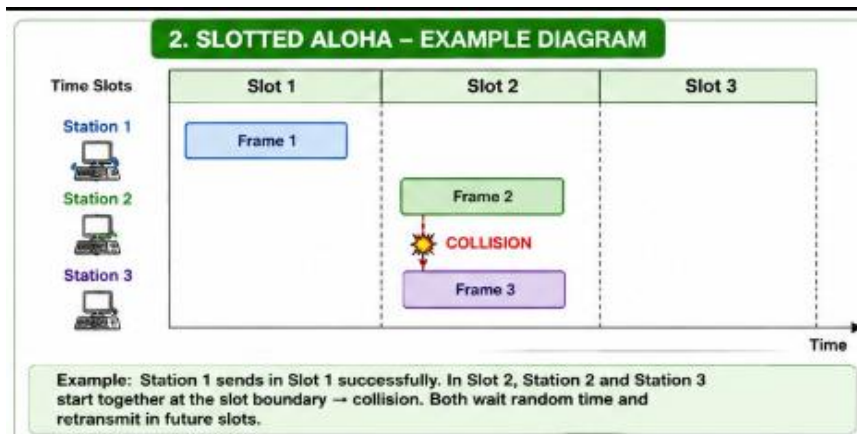
$$VT = T_{fr}$$

Throughput

$$S = G \times e^{-G}$$

Maximum efficiency $\approx 37\%$

Example



Category B: CSMA

CSMA = Carrier Sense Multiple Access

1. CSMA works on "**Sense before Transmit**" or "**Listen before Talk**".
2. A station first checks whether the channel is free or busy.
3. If the channel is free, it transmits.
4. If the channel is busy, it follows the appropriate waiting procedure.

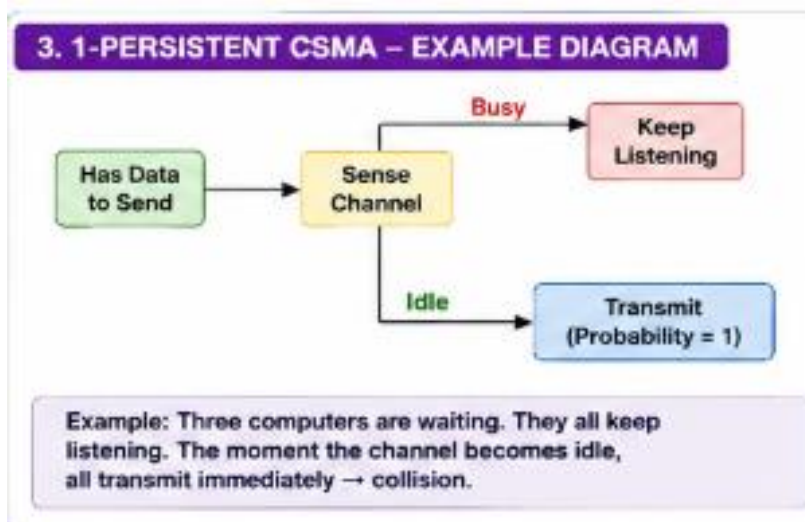
Types of CSMA

1. **1-Persistent CSMA**
2. **Non-Persistent CSMA**
3. **P-Persistent CSMA**

1. 1-Persistent CSMA

1. Station senses the channel.
2. If the channel is busy, it **continuously monitors** it.
3. When the channel becomes free, it transmits immediately.
4. It transmits with **probability 1 (100%)**.

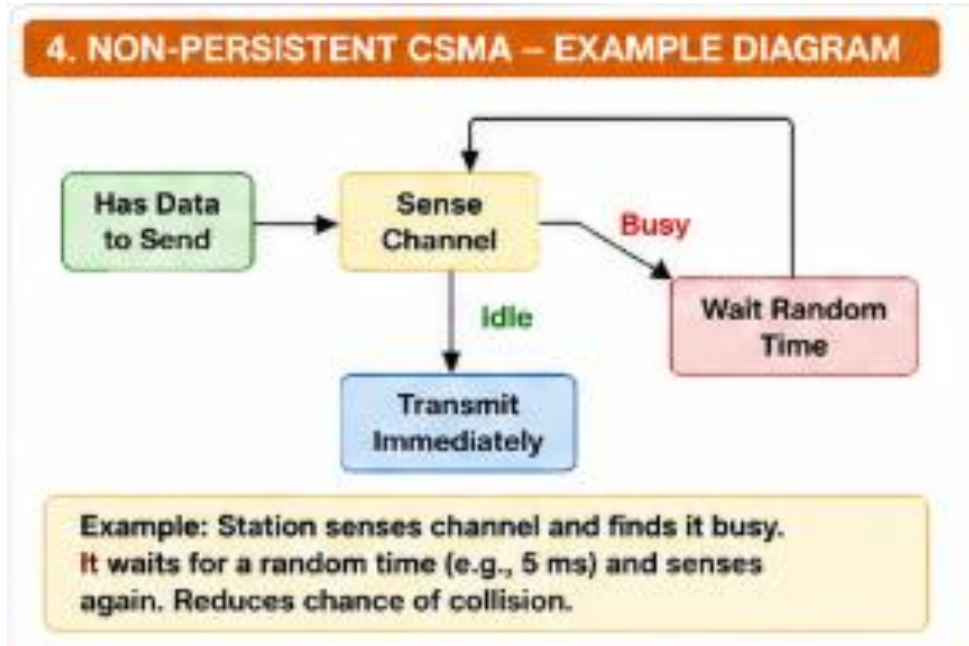
Example:



2. Non-Persistent CSMA

1. Station senses the channel.
2. If the channel is free, it transmits immediately.
3. If the channel is busy, it **does not continuously monitor** it.
4. It waits for a **random period** and senses the channel again.

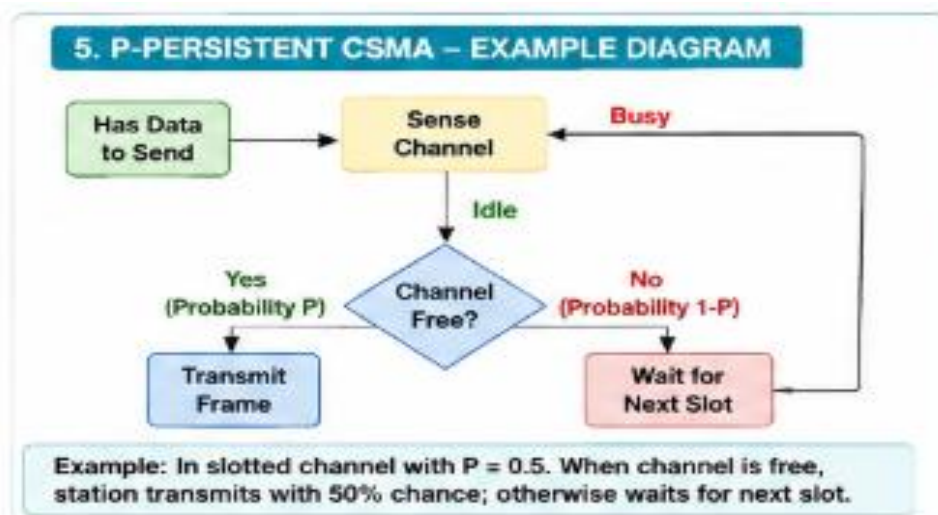
Example:



3. P-Persistent CSMA

1. It is used with **slotted channels**.
2. When the channel becomes free, the station transmits with probability **P**.
3. With probability **1 – P**, it waits for the next time slot.
4. It repeats the process until transmission occurs.

Example:

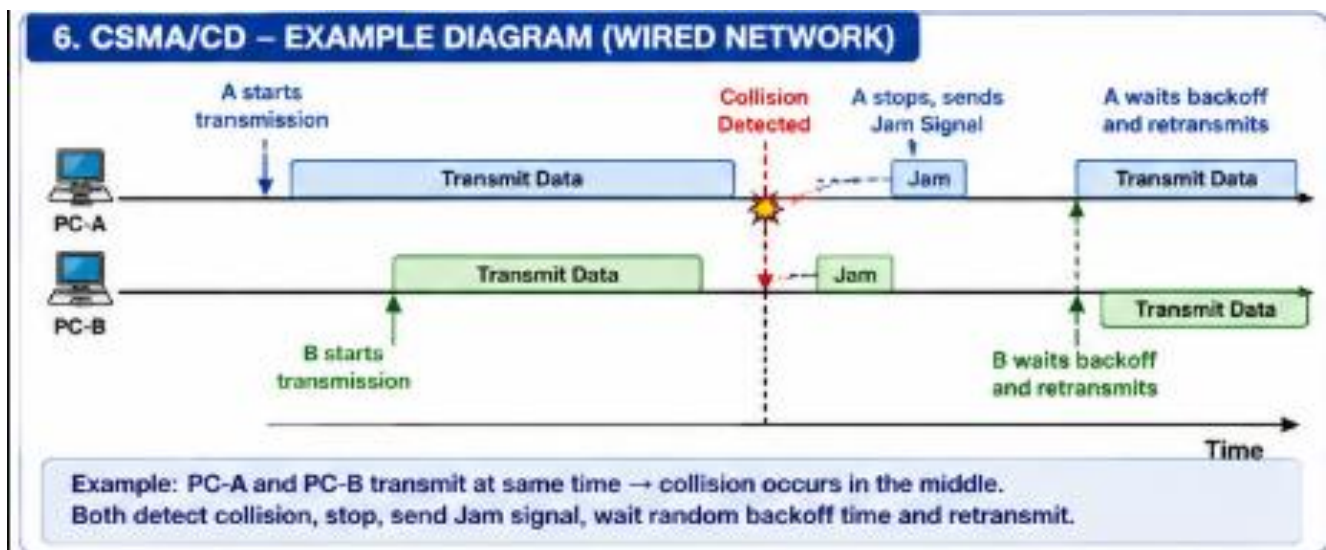


Category C: CSMA/CD

CSMA/CD = Carrier Sense Multiple Access with Collision Detection

1. Mainly used in **shared wired networks**.
2. It follows "**Listen before sending + Listen while sending**".
3. The station checks the channel before transmission.
4. It also monitors the channel during transmission.
5. If a collision occurs, it **stops transmission**.
6. It sends a **Jam Signal**.
7. It waits for a random **Back-off Time**.
8. Then it retransmits the frame.

Example

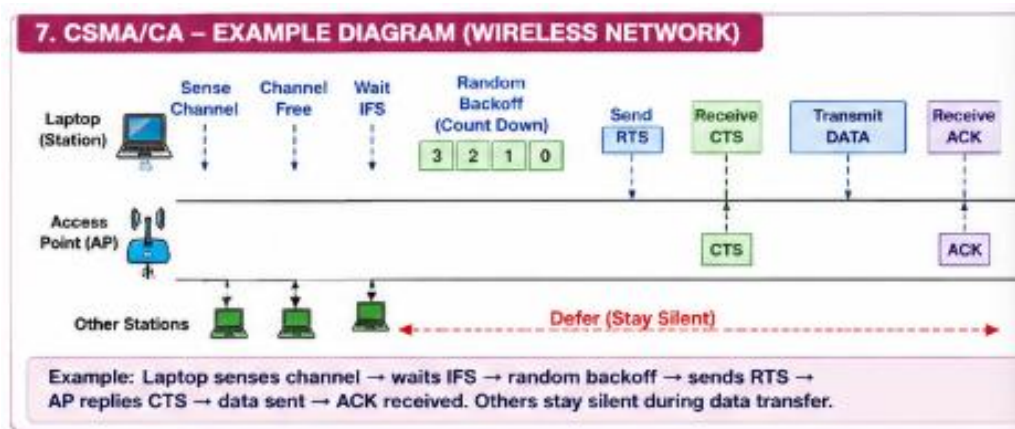


Category D: CSMA/CA

CSMA/CA = Carrier Sense Multiple Access with Collision Avoidance

1. Mainly used in **wireless networks (Wi-Fi)**.
2. Collision detection during wireless transmission is difficult.
3. It tries to **avoid collisions before they occur**.
4. It uses **Inter-Frame Space (IFS)** and a **Random Backoff Timer**.
5. It can also use **RTS/CTS** to reserve the channel.
6. The receiver sends an **ACK** after successful reception.

Example



Difference Between CSMA/CD and CSMA/CA

Feature	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Environment	Shared wired networks	Wireless networks
Main Strategy	Detects collision after it occurs	Tries to avoid collision before it occurs
Collision Detection	Yes	No
IFS	Not used	Used
Backoff	Applied after collision	Applied before transmission
Control Signal	Uses Jam Signal	Uses RTS/CTS
ACK	Not mandatory	Required
Example	Hub-based Ethernet	Wi-Fi (IEEE 802.11)